Chapter 4 Probability and Sampling Distribution

11 June 2024 00:00

- Random Variable :-

⇒ A random variable is a variable whose value is a numerical outcome of a random phenomina.

ex:- Let X ← random variable denote the number of heads

∴ if outcome is HTH ⇒ X = 2
THH ⇒ X = 1.

- Probability Distribution:

⇒ The probability distribution of a random variable X tells us what values X can take & how to assign probabilities to those values.

- Discrete random variables :-

⇒ If X takes on finite number of values then it's called discrete random variable.

- Discrete Probability Distribution :-

⇒ The probability distributions of a discrete random variable X lists the possible values of X & their probability.

Value of X :	x_1	x_2	x_3	⋯	x_k
Probability :	p_1	p_2	p_3	⋯	p_k

⇒ To find the probability of any event just add the probabilities p_i of the individual values x_i that make up that event.

EXAMPLE 4.9 Hard-Drive Sizes

Buyers of a laptop computer model may choose to purchase either a 120-, 160-, 250-, or 320-gigabyte (GB) internal hard drive. Choose customers from the last 60 days at random to ask what size internal hard drive they purchased for their computer. To "choose at random" means to give every customer of the last 60 days the same chance to be chosen. The size (in gigabytes) of the internal hard drive purchased by a randomly selected customer is a random variable X.

The value of X changes when we repeatedly choose customers at random, but it is always one of 120, 160, 250, or 320 GB. Here is the distribution of X:

Hard-drive size X:	120	160	250	320
Probability	0.50	0.25	0.15	0.10

The probability histogram in Figure 4.8 pictures this distribution.

The probability that a randomly selected customer purchased at least a 250-GB hard drive is

$$P(\text{size is 250 or 320}) = P(X = 250) + P(X = 320)$$
$$= 0.15 + 0.10 = 0.25$$

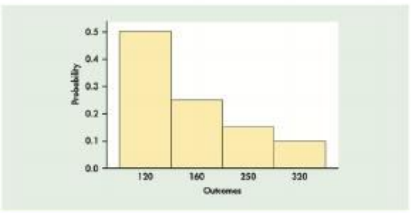
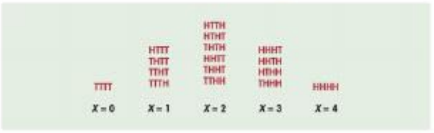


FIGURE 4.8 Probability histogram for the distribution of hard-drive sizes in Example 4.9.

- Example 4.10 : Four coin tosses:-

⇒ toss a balance coin four times; the discrete random variable X counts the number of heads.

FIGURE 4.9 Possible outcomes in four tosses of a coin, for Example 4.10. The random variable X is the number of heads.



The distribution of an unbiased coin toss is nearly symmetrical (gaussian).

⇒ Now,

$$P(X = 2) = \frac{\text{count of ways } X = 2 \text{ can occur}}{\text{total outcomes}}$$

$$= \frac{6}{16}.$$

⇒ In table form the entire distribution,

Number of heads X :	0	1	2	3	4
Probability :	0.0625	0.25	0.375	0.25	0.0625

⇒ Any event involving the number of heads observed can be expressed in terms of X, and its probability can be found from the distribution of X.

ex: the probability of atleast 2 heads.

$$P(X \geq 2) = 0.375 + 0.25 + 0.0625 = 0.6875.$$

⇒ The probability of atleast one head is most simply found by,

$$P(X \geq 1) = 1 - P(X = 0)$$
$$= 1 - 0.0625 = 0.9375.$$

- Continous random variable :-

⇒ A continous random variable has infinite number of possible values.

- Continous Probability Distributions:-

⇒ The probability distribution of a continous random variable X is described by a density curve.

→ The probabilities of any event is the area under the density curve and above the values of X that make up the event.

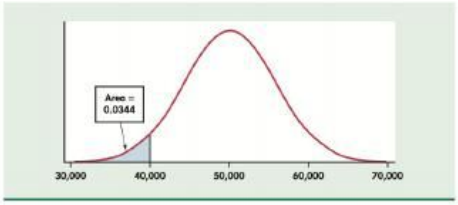
EXAMPLE 4.11 Tread Life

The actual tread life X of a 40,000-mile automobile tire has a Normal probability distribution with $\mu = 50{,}000$ miles and $\sigma = 5500$ miles. We say X has an $N(50{,}000, 5500)$ distribution. From a manufacturer's perspective, it would be useful to know the percent of tires that fail to meet the guaranteed wear life of 40,000 miles. This implies we want to find the probability that a randomly selected tire has a tread life less than 40,000 miles: z-score

$$P(X < 40{,}000) = P\left(\frac{X - 50{,}000}{5500} < \frac{40{,}000 - 50{,}000}{5500}\right)$$
$$= P(Z < -1.82)$$
$$= 0.0344$$

The $N(50{,}000, 5500)$ distribution is shown in Figure 4.11 with the area $P(X < 40{,}000)$ shaded. The manufacturer should expect to incur warranty costs for about 3.4% of its tires.

FIGURE 4.11 The Normal distribution with $\mu = 50{,}000$ and $\sigma = 5500$. The shaded area is $P(X < 40{,}000)$, calculated in Example 4.11.

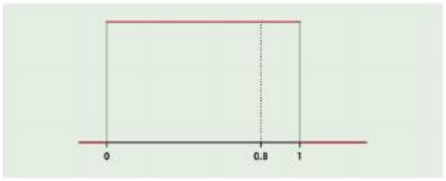


* The probability distribution for a continuous random variable assign probabilities to intervals of outcomes rather than individual outcomes.
  - All continuous probability distributions assign probability 0 to every individual outcome.

EXAMPLE 4.12 Can a Random Number Be Exactly 0.8?

What is $P(Y = 0.8)$, the probability that a random number generator produces *exactly* 0.8? Figure 4.12 shows the uniform density curve for outcomes between 0 and 1. Probabilities are areas under this curve. The probability of any interval of outcomes is equal to its length. This implies that because the point 0.8 has no length, its probability is 0.

FIGURE 4.12 The density curve of the uniform distribution of numbers chosen at random between 0 and 1, for Example 4.12. This model assigns probability 0 to the event that the number generated has any one specific value, such as 0.8.



Although this fact may seem odd, it makes intuitive, as well as mathematical, sense. The random number generator produces a number between 0.79 and 0.81 with probability 0.02. An outcome between 0.799 and 0.801 has probability 0.002. A result between 0.799999 and 0.800001 has probability 0.000002. You see that as we home in on 0.8 the probability gets closer to 0. To be consistent, the probability of an outcome *exactly* equal to 0.8 must be 0.

Because there is a zero probability exactly at $Y = 0.8$, the two events $\{Y > 0.8\}$ and $\{Y \geq 0.8\}$ have the same probability. We can ignore the distinction between $>$ and $\geq$ when finding probabilities for continuous (but not discrete) random variables.

o The mean of a random variable:

→ The mean of a random variable X is also an average of the possible values of X, just with an essential change to take into account the fact that not all outcomes are equally likely.

EXAMPLE 4.13 Pick 3

Most states and Canadian provinces have government-sponsored lotteries. Here is a simple lottery wager, from the Tri-State Pick 3 game that New Hampshire shares with Maine and Vermont. You choose a three-digit number; the state chooses a three-digit winning number at random and pays you $500 if your number is chosen. Because there are 1000 three-digit numbers, you have probability 1/1000 of winning. Taking X to be the amount your ticket pays you, the probability distribution of X is

Payoff X:	$0	$500
Probability:	0.999	0.001

~~What is your average payoff from many tickets?~~ The ordinary average of the two possible outcomes $0 and $500 is $250, but that makes no sense as the average because $500 is much less likely than $0. In the long run you receive $500 once in every 1000 tickets and $0 on the remaining 999 of 1000 tickets. The long-run average payoff is

$$\$500\frac{1}{1000} + \$0\frac{999}{1000} = \$0.50$$

Mean is good to estimate average pay off.

or fifty cents. That number is the mean of the random variable X. ~~(Tickets cost $1, so in the long run the state keeps half the money you wager.)~~

→ The mean of a random variable X is represented as μ_X.
  - It's also called the expected value of X, although we don't expect one observation on X to be close to its mean.

- Mean of a Discrete Random Variable:

→ Suppose that X is a discrete random variable whose distribution is

Value of X :	x_1	x_2	x_3	x_4	...	x_k
Probability :	p_1	p_2	p_3	p_4	...	p_k

→ To find mean of X, multiply each possible value by its probability, then add all the products.

$$\mu_X = x_1 p_1 + \dots + x_k p_k$$

$$\boxed{\mu_X = \sum x_i p_i}$$

EXAMPLE 4.14 First Digits

CASE 4.1 If first digits in a set of records appear "at random," the nine possible digits 1 to 9 all have the same probability. The probability distribution of the first digit X is then

First digit X:	1	2	3	4	5	6	7	8	9
Probability:	1/9	1/9	1/9	1/9	1/9	1/9	1/9	1/9	1/9

The mean of this distribution is

$$\mu_X = 1\left(\frac{1}{9}\right)+2\left(\frac{1}{9}\right)+3\left(\frac{1}{9}\right)+4\left(\frac{1}{9}\right)+5\left(\frac{1}{9}\right)+6\left(\frac{1}{9}\right)+7\left(\frac{1}{9}\right)+8\left(\frac{1}{9}\right)+9\left(\frac{1}{9}\right)$$
$$= 45 \times \frac{1}{9} = 5$$

- Rules for Means:

CASE 4.2 **Portfolio Analysis** The *rate of return* of an investment over a time period is the percent change in the price during the time period, plus any income received. For example, Apple Computer's stock price was $169.53 at the beginning of August 2008 and $113.66 at the end of that month. Apple pays no dividends, so Apple's rate of return for that time period was

$$\frac{\text{change in price}}{\text{starting price}} = \frac{113.66 - 169.53}{169.63} = -0.33, \text{ or } -33\%$$

Investors want high positive returns, but they also want safety. Over the course of a nearly 20-year period (1988 to 2008), Apple's monthly returns have swung to as low as −57% and to as high as +45%. The variability of returns, called *volatility* in finance, is a measure of the risk of an investment. A highly volatile stock, which may go either up or down a lot, is more risky than a Treasury bill, whose return is very predictable.

A *portfolio* is a collection of investments held by an individual or an institution. *Portfolio analysis* begins by studying how the risk and return of a portfolio are determined by the risk and return of the individual investments it contains. That's where statistics comes in: The returns on an investment over some period of time is a random variable. We are interested in the *mean* return and we measure volatility by the *standard deviation* of returns. Let's say that Sadie is interested in building a computer technology–based portfolio. Her intuition is that mixing a stock based on a Microsoft-based PC and an Apple stock will hedge the risk of one stock versus the other. Sadie decides to place 60% of her funds in Dell Inc. and 40% in Apple Inc. If X is the monthly return on Dell and Y the monthly return on Apple, the portfolio rate of return is

$$R = 0.6X + 0.4Y$$

How can we find the mean and standard deviation of the portfolio return R starting from information about X and Y? We must now develop the machinery to do this.



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- Rules for Means :-

→ Rule 1. If X is a random variable and a & b are fixed numbers then $\boxed{\mu_{a+bX} = a + b\mu_X}$

→ Rule 2: If X and Y are random variables, then

$$\boxed{\mu_{X+Y} = \mu_X + \mu_Y}$$

These are addition rules of mean.

- Returning example, example 4.15

CASE 4.2 DATA FILE PORTFOLIO

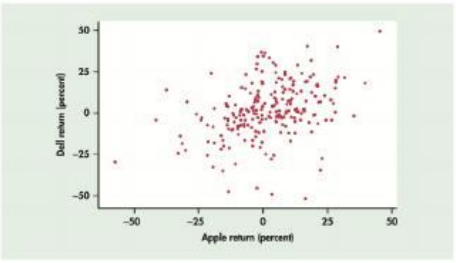
The past behavior of the two securities in Sadie's portfolio is pictured in Figure 4.14, which plots the monthly returns for Dell stocks against Apple stocks from January 1990 to August 2008. We can see that the returns on the two stocks are correlated. This fact will be used later for gaining a complete assessment of the expected performance of the expected performance of the portfolio. For now, we can calculate mean returns from the 224 data points shown on the plot:[10]

X = monthly return for Dell $\quad \mu_X = 2.02\%$
Y = monthly return for Apple $\quad \mu_Y = 1.75\%$

Sadie invests 60% in Dell and 40% in Apple. We can find the mean return on her portfolio by combining Rules 1 and 2:

$$R = 0.6X + 0.4Y$$
$$\mu_R = 0.6\mu_X + 0.4\mu_Y$$
$$= (0.6)(2.02) + (0.4)(1.75) = 1.91\%$$

FIGURE 4.14 Monthly returns on Dell stock versus returns on Apple stock (January 1990 to August 2008) for Example 4.15.



- **Variance of a Discrete Random Variable:**

⇒ Suppose that X is a random variable whose distribution is

Value of X :	x_1	x_2	x_3	.....	x_k
Probability :	p_1	p_2	p_3	.....	p_k

and that μ_x is the mean of x. Then the variance of X

$$\sigma_x^2 = (x_1 - \mu_x)^2 p_1 + (x_2 - \mu_x)^2 p_2 + \dots + (x_k - \mu_x)^2 p_k$$

$$\boxed{\sigma_x^2 = \sum (x_i - \mu_x)^2 p_i}.$$

⇒ **The standard deviation** of X is the square root of the variance.

- **The rules of Variance :-**

Rule 1: If X is a random variable & a, and b are fixed numbers, then

$$\sigma^2_{a+bx} = b^2 \sigma^2_x \quad \text{(the constant is removed)}$$

Rule 2: If X and Y are **independent** random variables, then

$$\sigma^2_{x+y} = \sigma^2_x + \sigma^2_y$$
$$\sigma^2_{x-y} = \sigma^2_x + \sigma^2_y$$

⇒ This is the **addition rule for variances of independent random variable.**

Rule 3: If X and Y have correlation ρ, then

$$\sigma^2_{x+y} = \sigma^2_x + \sigma^2_y + 2\rho\sigma_x\sigma_y.$$
$$\sigma^2_{x-y} = \sigma^2_x + \sigma^2_y - 2\rho\sigma_x\sigma_y.$$

⇒ This is the general addition rule for variances of random variable.

✱ Variance of independent random variable add ; standard deviations do ✱✱ not.

EXAMPLE 4.18 Pick 3

The payoff X of a \$1 ticket in the Tri-State Pick 3 game is \$500 with probability 1/1000 and 0 the rest of the time. Here is the combined calculation of mean and variance:

x_i	p_i	$x_i p_i$	$(x_i - \mu_X)^2 p_i$
0	0.999	0	$(0 - 0.5)^2(0.999) = 0.24975$
500	0.001	0.5	$(500 - 0.5)^2(0.001) = 249.50025$
		$\mu_X = 0.5$	$\sigma_X^2 = 249.75$

The standard deviation is $\sigma_X = \sqrt{249.75} = \$15.80$. Games of chance usually have large standard deviations because large variability makes gambling exciting.

If you buy a Pick 3 ticket, your net winnings are $W = X - 1$ because the dollar you pay for the ticket must be subtracted from the payoff. By the rules for means, the mean amount you win is

$$\mu_W = \mu_X - 1 = -\$0.50$$

That is, you lose an average of 50 cents on a ticket. The rules for variances remind us that the variance and standard deviation of the winnings $W = X - 1$ are the same as those of the payoff X. Subtracting a fixed number changes the mean but not the variance.

Suppose now that you buy a \$1 ticket on each of two different days. The payoffs X and Y on the two tickets are independent because separate drawings are held each day. Your total payoff $X + Y$ has mean

$$\mu_{X+Y} = \mu_X + \mu_Y = \$0.50 + \$0.50 = \$1.00$$

Because X and Y are independent, the variance of $X + Y$ is

$$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 = 249.75 + 249.75 = 499.5$$

The standard deviation of the total payoff is

$$\sigma_{X+Y} = \sqrt{499.5} = \$22.35$$

This is not the same as the sum of the individual standard deviations, which is \$15.80 + \$15.80 = \$31.60. Variances of independent random variables add; standard deviations do not.

EXAMPLE 4.20 Portfolio Analysis

CASE 4.2 Now we can complete our initial analysis of Sadie's portfolio. Based on monthly returns between 1990 and 2008, we have

X = monthly return for Dell $\quad \mu_X = 2.02\% \quad \sigma_X = 16.59\%$
Y = monthly return for Apple $\quad \mu_Y = 1.75\% \quad \sigma_Y = 15.09\%$
Correlation between X and Y: $\quad \rho = 0.379$

We see that Dell has a higher mean return than Apple but is also more volatile. This is not an uncommon relationship in finance: a more volatile investment often offers a higher return, compensating investors for its greater risk. For the return R on Sadie's portfolio of 60% for Dell and 40% for Apple,

$$R = 0.6X + 0.4Y$$
$$\mu_R = 0.6\mu_X + 0.4\mu_Y$$
$$= (0.6 \times 2.02) + (0.4 \times 1.75) = 1.91\%$$

↳ Investment gave higher return.

To find the variance of the portfolio return, combine Rules 1 and 3:

$$\sigma_R^2 = \sigma_{0.6X}^2 + \sigma_{0.4Y}^2 + 2\rho\sigma_{0.6X}\sigma_{0.4Y}$$
$$= (0.6)^2\sigma_X^2 + (0.4)^2\sigma_Y^2 + 2\rho(0.6 \times \sigma_X)(0.4 \times \sigma_Y)$$
$$= (0.6)^2(16.59)^2 + (0.4)^2(15.09)^2 + (2)(0.379)(0.6 \times 16.59)(0.4 \times 15.09)$$
$$= 181.06$$
$$\sigma_R = \sqrt{181.06} = 13.46\%$$

↳ Investment is less volatile than apple

The portfolio has a smaller mean return than investing in all Dell stocks would, but it is also less volatile. As a proportion of the Dell values, the reduction in standard deviation is greater than the reduction in mean return. That's why Sadie put some funds into Apple stock.

⇒ The above example also suggest the characteristic weakness of portfolio analysis: risk & return in the future may be very different from the σ and μ we get from the past returns.

◦ The sampling Distribution of a Sample Mean:

⇒ A statistics from a random sample will take different value if we take more samples from the same population.

↳ i.e Simple statistics are random variable.

EXAMPLE 4.21 Does this wine smell bad?

Sulfur compounds such as dimethyl sulfide (DMS) are sometimes present in wine. DMS causes

"off-odors" in wine, which from the winemakers' perspective is an important product quality attribute to monitor and control. Too high a DMS level can result in loss of sales. On the other hand, there is a cost for lowering DMS levels. It makes good business sense for winemakers to study the odor threshold, the lowest concentration of DMS that the human nose can detect. Different people have different thresholds, so we start by asking about the mean threshold μ in the population of all adults. The number μ is a *parameter* that describes this population.

To estimate μ, we present tasters with both natural wine and the same wine spiked with DMS at different concentrations to find the lowest concentration at which they can identify the spiked wine. Here are the odor thresholds (measured in micrograms of DMS per liter of wine) for 10 randomly chosen subjects:

28 40 28 33 20 31 29 27 17 21

The mean threshold for these subjects is $\bar{x} = 27.4$. This sample mean is a *statistic* that we use to estimate the parameter μ, but it is probably not exactly equal to μ. ~~Moreover, we know that a different 10 subjects would give us a different $\bar{x}$.~~

- Statistical estimation & the law of large numbers :-

Q) If $\bar{x}$ is rarely exactly right to the population mean & varies samples from samples, why is it nonetheless a reasonable estimate of the population mean μ?

A) If we keep on taking larger & larger samples, the statistic $\bar{x}$ is guaranteed to get closer & closer to μ, because variance reduces.

→ This holds for any population, not just some special case such as normal distribution.

- Law of Large Numbers :

→ Draw independent observations at random from any population with finite mean μ. As the number of observations drawn increases, the mean $\bar{x}$ of the observed values gets closer & closer to the mean μ of the population.

EXAMPLE 4.22 The Law of Large Numbers in Action

In fact, the distribution of DMS odor thresholds among all adults has mean 25. The mean $\mu = 25$ is the true value of the parameter we seek to estimate. Typically, the value of a parameter like μ is unknown. We use an example in which μ is known to illustrate facts about the general behavior of $\bar{x}$ that hold even when μ is unknown. Figure 4.15 shows how the sample mean $\bar{x}$ of an SRS drawn from this population changes as we add more subjects to our sample.

The first subject in Example 4.21 had threshold 28, so the line in Figure 4.15 starts there. The mean for the first two subjects is

$$\bar{x} = \frac{28 + 40}{2} = 34$$

This is the second point on the graph. At first, the graph shows that the mean of the sample changes as we take more observations. Eventually, however, the mean of the observations gets close to the population mean $\mu = 25$ and settles down at that value.

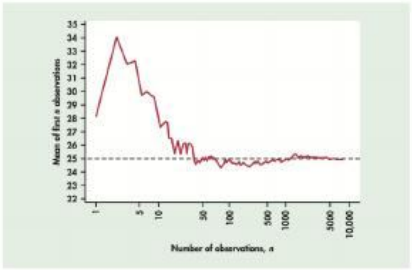


FIGURE 4.15 The law of large numbers in action: as we take more observations, the sample mean $\bar{x}$ always approaches the mean (μ) of the population.

- The "law of small numbers" :-

→ Most people believe in incorrect "law of small numbers". ↳ something that does exist.

→ That is, we expect even short sequence of random events to show the kind of average behaviour that in fact appears only in the long run.

- How large is a large number?

→ law of large number doesn't say how many trials/samples are needed

to guarantee a mean outcome close to μ (Population mean)

↳ That depends on variability, the more variable the trials the more trials are needed to ensure that the mean outcome $\bar{x}$ is closer to μ.

Is there a winning system for gambling?

Serious gamblers often follow a system of betting in which the amount bet on each play depends on the outcome of previous plays. You might, for example, double your bet on each spin of the roulette wheel until you win—or, of course, until your fortune is exhausted. Such a system tries to take advantage of the fact that you have a memory even though the roulette wheel does not. Can you beat the odds with a system based on the outcomes of past plays? No. Mathematicians have established a stronger version of the law of large numbers that says that if you do not have an infinite fortune to gamble with, your long-run average winnings μ remain the same as long as successive trials of the game (such as spins of the roulette wheel) are independent.

What if observations are not independent?

You are in charge of a process that manufactures video screens for computer monitors. Your equipment measures the tension on the metal mesh that lies behind each screen and is critical to its image quality. You want to estimate the mean tension μ for the process by the average $\bar{x}$ of the measurements. Alas, the tension measurements are not independent. If the tension on one screen is a bit too high, the tension on the next is more likely to also be high. Many real-world processes are like this—the process stays stable in the long run, but observations made close together are likely to be both above or both below the long-run mean. ~~Again the mathematicians come to the rescue: as long as the dependence dies out fast enough as we take measurements farther and farther apart in time, the law of large numbers still holds.~~

⇒ ∵ only the closer observation has dependence in previous example, & the dependence doesn't hold far enough the law of large number still hold.

- The mean & the standard deviation of sample mean $\bar{x}$

⇒ Here $\bar{x}$ is the random variable (having sampling distribution)
↳ $\bar{x}$ changes values when different sets of samples are taken.

Suppose that $\bar{x}$ (still a random variable.) is the mean of an SRS of size n drawn from a large population with mean μ and the standard deviation σ.

↳ Then the mean of the sampling distribution of $\bar{x}$ is μ and its standard deviation is $\sigma/\sqrt{n}$.

↳ Note how it's $\frac{\text{population std}}{\sqrt{n}}$,

Unlike population mean = mean of sampling distribution.

⇒ Both mean & standard deviation of the sampling distribution of $\bar{x}$ have important implications for statistical inference :—

1) The mean of the statistic $\bar{x}$ is always the same as the mean μ of the population.

↳ Because there is no systematic tendency (or favoritism) to underestimate or overestimate the parameter.

↳ This makes the idea of lack of bias more precise, and hence we say that the statistic $\bar{x}$ is an unbiased estimator of the parameter μ.

2) An unbiased estimator is "correct on the average" in many samples.

↳ How close the estimator falls to the parameter is determined by the spread of sampling distribution $\frac{\sigma}{\sqrt{n}}$

↳ ∵ the std gets smaller only by a factor of $\sqrt{n}$, to reduce the variance by 1/2, you need 4 times the sample.

for population with normal distribution

- Sampling Distribution of the sample mean.

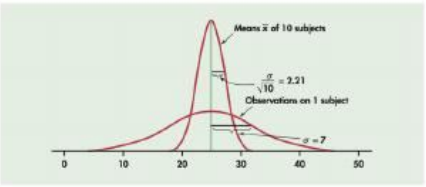
⇒ If a population has the $N(\mu, \sigma)$ distribution, then the sample mean $\bar{x}$ of n independent observations has the $N(\mu, \frac{\sigma}{\sqrt{n}})$ distribution.

↳ i.e. if the population distribution is normal, then so is the distribution of sample mean.

EXAMPLE 4.23 Estimating Odor Thresholds

Adults differ in the smallest amount of DMS they can detect in wine. Extensive studies have found that the DMS odor threshold of adults follows roughly a Normal distribution with mean $\mu = 25$ micrograms per liter ($\mu g/l$) and standard deviation $\sigma = 7\ \mu g/l$. Because the population distribution is Normal, the sampling distribution of $\bar{x}$ is also Normal. Figure 4.16 displays the

FIGURE 4.16 The distribution of single observations compared with the distribution of the means $\bar{x}$ of 10 observations, for Example 4.23. Averages are less variable than individual observations.



- Central limit theorem:

⇒ Draw an SRS of size n from any population with mean μ & finite std σ.

↳ When n is large, the sampling distribution (distribution of) the sample mean $\bar{x}$ is approximately Normal.

$\therefore \bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ (Sampling distribution of the mean is approximately normal)

regardless of the population.

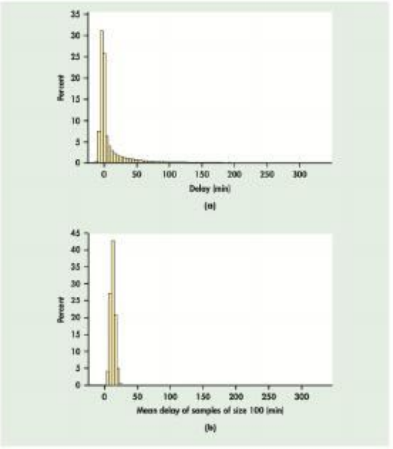
EXAMPLE 4.24 The Central Limit Theorem in Action

DATA FILE UNITED

In 2009, there were 73,423 departures for United Airlines from its largest hub airport, O'Hare Airport in Chicago. Figure 4.17(a) is a histogram of the departure delay times (in minutes) for the entire population of United flights from O'Hare.[24] A negative departure delay represents a flight that left earlier than its scheduled departure time. The distribution is clearly very different from the Normal distribution. It is strongly skewed to the right and very spread out. The right tail of the distribution is even longer than what appears in the figure because there are too few high delay times for the histogram bars to be visible on this scale. In fact, we cut off the departure delay time scale at 350 minutes to save space.

The mean μ of the population is 11.793 min. Suppose we take an SRS of 100 flights. The mean delay time in this sample is $\bar{x} = 8.09$ min. That's less than the mean of the population. Take another SRS of size 100. The mean for this sample is $\bar{x} = 17.37$ min. That's higher than the mean of the population. If we take more samples of size 100, we will get different values of $\bar{x}$. To find the sampling distribution of $\bar{x}$, we take many random samples of size 100 and calculate $\bar{x}$ for each sample. Figure 4.17(b) is a histogram of the mean departure delay times for 1000 samples, each of size 100. Notice something remarkable. Even though the distribution of the individual delay times

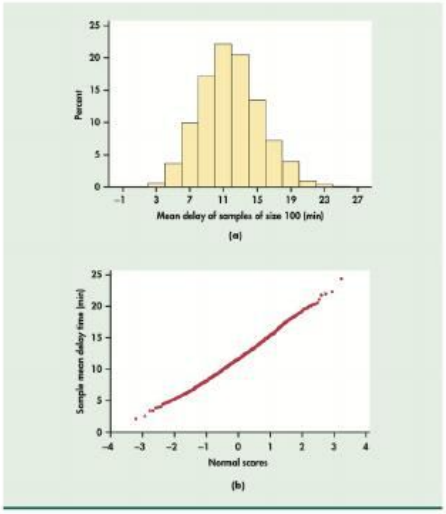
distribution of departure delay times in a population of 79,423 departures, for Example 4.24. (b) The distribution of sample means $\bar{x}$ for 1000 random samples of size 100 from this population. Both histograms have the same scales and histogram classes to allow for direct comparison.



is strongly skewed and very spread out, the distribution of the sample means is quite symmetric and much less spread out.

Figure 4.18(a) is the histogram of the sample means on a scale that more clearly shows its shape. We can see that the distribution of sample means is close to the Normal distribution. The Normal quantile plot of Figure 4.18(b) further confirms the compatibility of the distribution of sample means with the Normal distribution. Furthermore, as expected, the histogram in Figure 4.18(a) appears to be essentially centered on the population mean μ value. Specifically, the mean of the 1000 sample means is 11.831, which is nearly equal to the μ-value of 11.793. If more and more random samples are taken, the mean of the sample mean values will eventually equal μ exactly.

FIGURE 4.18 (a) The distribution of sample means $\bar{x}$ from Figure 4.17(b) shown in more detail. (b) Normal quantile plot of the 1000 sample means. The distribution is close to Normal.



EXAMPLE 4.26 Maintaining Air Conditioners

The time X that a technician requires to perform preventive maintenance on an air-conditioning unit is governed by the exponential distribution whose density curve appears in Figure 4.19(a). The mean time is $\mu = 1$ hour and the standard deviation is $\sigma = 1$ hour. Your company operates 70 of these units. What is the probability that their average maintenance time exceeds 50 minutes?

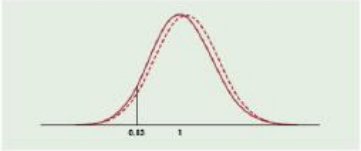
The central limit theorem says that the sample mean time $\bar{x}$ (in hours) spent working on 70 units has approximately the Normal distribution with mean equal to the population mean $\mu = 1$ hour and standard deviation

$$\frac{\sigma}{\sqrt{70}} = \frac{1}{\sqrt{70}} = 0.12 \text{ hour}$$

The distribution of $\bar{x}$ is therefore approximately $N(1, 0.12)$. This Normal curve is the solid curve in Figure 4.20.

average maintainence per unit
can be treated as taking
sample distribution of size
70 units (per observation)
→ we have access to population
mean

FIGURE 4.20 The exact distribution (dashed) and the Normal approximation from the central limit theorem (solid) for the average time needed to maintain an air conditioner, for Example 4.26.



Because 50 minutes is 50/60 of an hour, or 0.83 hour, the probability we want is $P(\bar{x} > 0.83)$. A Normal distribution calculation gives this probability as 0.9222. This is the area to the right of 0.83 under the solid Normal curve in Figure 4.20.

Using more mathematics, we could start with the exponential distribution and find the actual density curve of $\bar{x}$ for 70 observations. This is the dashed curve in Figure 4.20. You can see that the solid Normal curve is a good approximation. The exactly correct probability is the area under the dashed density curve. It is 0.9294. The central limit theorem Normal approximation is off by only about 0.007.